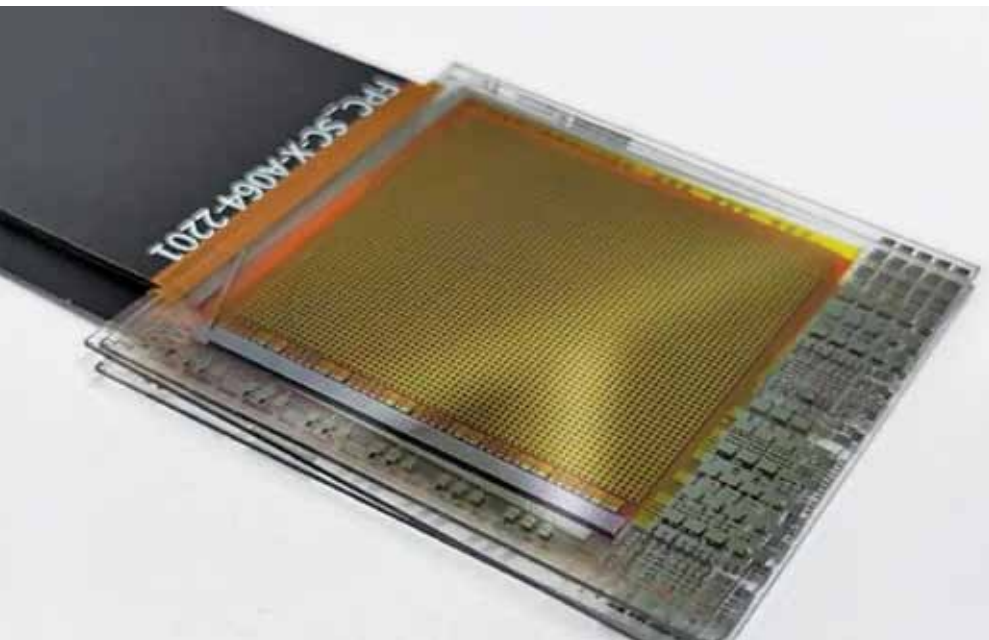




RESEARCH & INNOVATION UPDATES

Perovskite image sensors capture 100% light



ETH Zurich's (Public university in Zürich, Switzerland) team has created stacked perovskite image sensors that absorb red, green, and blue light directly, with no filters needed. Using chemically tuned layers, this design captures 100% of incoming light, tripling both resolution and efficiency. Early thin-film prototypes hint at significant potential for hyper-spectral imaging in medicine, agriculture, and other applications. Future miniaturisation could make this ready for next-gen cameras and diagnostics.

One of the two perovskite-based sensor prototypes that the researchers have used to demonstrate that the technology can be successfully miniaturised (Credit: Empa/ETH Zurich)

3D smart pen spots early signs of Parkinson's

A team at UCLA (Los Angeles, California) has developed a self-powered diagnostic pen that can detect early signs of Parkinson's disease with over 96% accuracy by analysing handwriting movements. The pen uses a soft silicone tip with magnetic ink and a coil of conductive yarn that generates electrical signals as the user writes; no battery required. These signals capture subtle motor impairments, often invisible to the eye, and are processed by a neural network trained to recognise early Parkinson's symptoms. Unlike costly scans or specialist assessments, this portable, low-cost tool can be used during routine check-ups, making it ideal for low-resource settings.



The highly sensitive diagnostic pen features a soft, silicon magnetoelastic tip and ferrofluid ink—a special liquid containing tiny magnetic particles (Credit: <https://newsroom.ucla.edu>)